Primitive Divisors as First Returns in Congruence Dynamics

Route-A structural certificate C179 — round 0

Abstract

For coprime integers $a > b \geq 1$, multiplication by ab^{-1} on the finite unit group $(\mathbb{Z}/N\mathbb{Z})^{\times}$ defines an arithmetic dynamical system without fitted data. We prove that a prime p is a primitive divisor of $a^n - b^n$ exactly when the marked point 1 first returns at time n on the p-fiber. Classical Zsigmondy existence then supplies a return prime except for its two exact exception forms. That existence theorem is explicitly attributed and is not claimed new. This first-round note freezes the object, clock, equivalence, attribution, and scope before the prime-power and global layers are added.

Keywords: primitive divisor; first return; congruence dynamics; multiplicative order; Zsigmondy theorem.

中文摘要

对互素整数 $a > b \geq 1$，在有限单位群 $(\mathbb{Z}/N\mathbb{Z})^{\times}$ 上乘以 ab^{-1} 给出一个无需拟合数据的算术动力系统。本文证明：素数 p 是 $a^n - b^n$ 的本原素因子，当且仅当标记点 1 在模 p 纤维上首次于时刻 n 返回。经典的 Zsigmondy 定理给出除两个精确例外类型之外的存在性； 本文明确归属该经典结果，并不声称其为新定理。本轮先冻结对象、时钟、等价关系、 文献边界和适用范围，素数幂与全局层将在后续轮次加入。

1 Frozen arithmetic dynamics

Fix coprime integers $a > b \geq 1$. For every $N \geq 2$ with $(N, ab) = 1$, set

$$U_N = (\mathbb{Z}/N\mathbb{Z})^{\times}, \qquad q_N = ab^{-1} \pmod N, \qquad R_N(x) = q_N x,$$

and mark $1 \in U_N$. One multiplication is one discrete source step. No logarithmic prime roof, prime weight, target table, or fitted parameter is introduced.

2 Primitive-return theorem

Theorem. For every prime p and $n \geq 2$, the following are equivalent:

1. $p \mid a^n - b^n$ and $p \nmid a^m - b^m$ for $1 \leq m < n$;
2. $p \nmid ab$ and $\mathrm{ord}_p(ab^{-1}) = n$;
3. 1 has least positive return n under R_p.

Moreover, such a prime exists except when $(a, b, n) = (2, 1, 6)$, or when $n = 2$ and $a + b$ is a power of two.

Proof. A prime divisor of $a^n - b^n$ cannot divide ab. Thus $a^m \equiv b^m \pmod p$ is equivalent to $(ab^{-1})^m \equiv 1 \pmod p$. Excluding all earlier m is exactly the definition of multiplicative order, and the orbit of the mark is $1, q_p, q_p^2, \ldots$. The existence and exception sentence is the classical Zsigmondy theorem [1]; no new proof or strengthening of that theorem is claimed. □

Every primitive return prime for $n \geq 2$ is odd. Indeed, if a, b are odd, then $2 \mid a - b$ already at time one; if they have opposite parity, every $a^n - b^n$ is odd.

3 Evidence and provisional Route-A boundary

The deterministic ledger covers 63 coprime pairs through $a = 14$ and times through 10. These rows are regression sentinels only; the proof above has no cutoff. The prime first-return relation is an intrinsic arithmetic origin and therefore improves on a nonarithmetic control. It remains weak: a prime is presently a return *modulus*, not a selected global primitive orbit, and the source clock is not $\log p$. No target divisor, functional equation, counting law, or Hilbert–Pólya operator is claimed.

Scope. No target zero or prime table, local arithmetic factor, root number, automorphy input, prime-weighted global product, logarithmic prime roof, or Route-B authorization enters this note. Scope literal: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

No human, animal, clinical, personal, private, or sensitive data are used. No external funding or conflict of interest is reported. Package-local code and exact evidence accompany the paper. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer.

References

[1] K. Zsigmondy, "Zur Theorie der Potenzreste," *Monatshefte für Mathematik und Physik* 3 (1892), 265–284. doi:10.1007/BF01692444.